

Chris Rosales

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PROJECTS

BeatSight - AI-Powered Drum Transcription Platform

github.com/rosacry/BeatSight | June 2025 - Present

- Trained a 7.1M-parameter PyTorch CNN ensemble on 16.9M audio samples to transcribe songs into playable drum beatmaps; reached 94.55% balanced accuracy (F1: 0.9153) across 12 drum classes using a dual-model split -- one model for body drums on clean audio, one for cymbals on Demucs-separated stems.
- Custom CNN uses Coordinate Attention, Stochastic Depth, Gradient Centralization, and multi-task heads for simultaneous hit classification, velocity, and articulation; cleaned 1.17M mislabeled training samples via a confidence-scoring pipeline built on the Phase 1 model's own outputs.
- Built the complete production stack: .NET 8 desktop client (2D/3D/manuscript views), FastAPI backend (84% test coverage, 3,035 tests), React 18 TypeScript frontend, PostgreSQL, Redis -- live at beatsight.io on AWS S3, Modal GPU compute, Railway, and Cloudflare Pages.
- Integrated Demucs source separation, spectral flux onset detection, and pitch-preserved tempo adjustment (50-200%) so users can slow down or speed up a song without changing its pitch while practicing.

Raptors Voting Analysis - Election Data Platform

github.com/khandaker-abid/raptors-voting-analysis | August 2025 - December 2025

- Led development of a full-stack election data platform processing 26,636+ records across 11 MongoDB collections; features include geographic choropleth visualizations, Voting Rights Act ecological inference modeling, and preclearance jurisdiction analysis.
- Built a 21-stage ETL pipeline with 100% pass rate on 37 integration tests, pulling from Census Bureau, EAC EAVS, and MIT Election Lab via Spring Boot REST API (30+ endpoints across 6 controllers).

Mapperatorinator - Open-Source AI Beatmap Generator (Contributor)

github.com/OliBomby/Mapperatorinator | October 2025 - Present

- Contributing to a 219M-parameter Whisper-based encoder-decoder transformer that generates rhythm game beatmaps from mel-spectrograms -- classifier-free guidance, LoRA fine-tuning, 90% window overlap for long-form generation, diffusion-based coordinate refinement.
- Wrote 6,000+ lines across 11 features: batch queue system, AcoustID/Shazam audio fingerprinting, bf16 auto-detection (40-60% speedup on compatible hardware), and 7 bug fixes covering process termination, Hydra config parsing, and pywebview compatibility.

SSBU Adaptive AI - Adaptive Skyline Plugin for Smash Bros. Ultimate

github.com/rosacry/ssbu-adaptive-ai | March 2026 - Present

- Built a Rust workspace (92K+ lines, 546 tests) for a Skyline plugin that hooks Smash Bros. Ultimate's AI callbacks at the AArch64 instruction level via A64HookFunction, cross-compiled for Nintendo Switch.
- Designed and implemented a decay-weighted opponent habit learner (per-matchup profiles, 14-day half-life), human-realism engine (reaction windows 8-40 frames by cue complexity, commitment windows, execution noise), and a persona-fitting system for player-style imitation.
- Diagnosed and fixed a multi-build freeze: root cause was two NRO instances racing to call A64HookFunction on the same target address; resolved with multi-phase hash-derived jitter serialization.

EDUCATION

Bachelor of Science, Computer Science

Stony Brook University | Stony Brook, NY | Expected December 2026

EXPERIENCE

Camp Leader and Instructor

January 2020 - August 2022

Code Ninjas | Clifton Park, NY

- Taught Python and Java to 50+ students per session; mentored incoming instructors on curriculum and classroom management.

INVOLVEMENT

Board Member - Cybersecurity Club

September 2023 - Present

Stony Brook University

- Organized CTF competitions and security workshops for club members.

Board Member - Competitive Programming Club

September 2023 - Present

Stony Brook University

- Ran weekly problem-solving sessions for 50+ students covering algorithms and data structures; coordinated team entries for ICPC-style competitions.

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript, C#, Java, C++, C, SQL, Rust

Machine Learning & AI: PyTorch, deep learning, convolutional neural networks (CNN), neural network training, LLMs, NLP, audio signal processing, librosa, NumPy, Pandas, scikit-learn, Weights & Biases, Demucs, Whisper

Frameworks & Libraries: FastAPI, React 18, .NET 8, Spring Boot, Node.js, REST API development, microservices

Infrastructure & Tools: PostgreSQL, MongoDB, Redis, Docker, AWS S3, Modal, Railway, Cloudflare, Git, Linux, CI/CD, pytest, Playwright, unit testing, integration testing